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How disclosure medium affects investor reactions to CEO bragging, modesty, and humblebragging☆

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ABSTRACT

We examine if investor expectations of two common disclosure mediums (conference calls and Twitter) interact with a CEO's communication style to influence investor judgments. Consistent with theory, results show that when the disclosure medium is a conference call, investors are less willing to invest when the CEO is modest about positive firm performance compared to when the CEO brags. In contrast, when the disclosure medium is Twitter, investors are less willing to invest when the CEO brags about positive firm performance compared to when the CEO is modest. Further analysis reveals that perceived CEO credibility mediates the influence of a CEO's communication style and disclosure medium on investor judgments. Additionally, we find that regardless of the disclosure medium, investors are less willing to invest in a firm when the CEO humblebrags about positive firm performance relative to when he brags or is modest. Our study contributes to the emerging literature on social media and disclosures, and to the literature investigating how style features of disclosures influence investor judgments. Our results also have practical implications for firms and managers developing communication strategies for new disclosure mediums like Twitter.

1. Introduction

Investors regularly access social media to obtain financial information (Brunswick, 2014). To help meet this demand, CEOs and firms are increasingly disclosing financial information on social media (e.g., by using Twitter). It is now common practice for firms to tweet highlights from press releases or live tweet quotes from conference calls (Joyce, 2013). The explosive use of social media to disclose financial information raises an interesting question: Do investor reactions to a CEO's communication style depend on whether the interaction is on a traditional disclosure medium (like conference calls) or a new disclosure medium (like Twitter; Miller & Skinner, 2015)? We help answer this question by examining if investor expectations of conference calls and Twitter interact with a CEO's communication style to influence investor judgments. Specifically, we examine if investor reactions to a CEO's use of bragging, modesty, or humblebragging to describe positive firm performance depend on whether the disclosure medium is a conference call or Twitter.

Recent research suggests that style features of disclosures influence investor judgments and decisions (e.g., Hales, Kuang, & Venkataraman, 2011; Libby & Emett, 2014; Rennekamp, 2012; Tan, Wang, & Zhou, 2014). We examine bragging, modesty, and humblebragging because they are common communication styles used to influence how individuals perceive a presenter (Leary & Kowalski, 1990; Marwick & Boyd, 2011; Smith, 2010; Tice, Butler, Muraven, & Stillwell, 1995; Wosinka, Dabul, Whetstone-Dion, & Cialdini, 1996). Examining these communication styles is important because bragging and modesty are present to varying degrees in almost every firm or CEO disclosure that contains more than mere facts, while humblebragging is becoming increasingly popular, especially on Twitter (Sezer, Gino, & Norton, 2018; Wittels, 2012). Bragging refers to attributing success to internal factors such as one's own ability or effort (Wosinka et al., 1996; Zuckerman, 1979) and is typically accompanied by intensifying words (e.g., 'innovative,' 'state-of-the-art,' 'gigantic,' and 'slash; ' Miller, Cooke, Tsang,

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& Morgan, 1992).[1] Conversely, modesty refers to attributing success to external factors such as luck or assistance from others (Carlston & Shovar, 1983; Miller et al., 1992; Tice et al., 1995) and typically lacks intensifying language (Miller et al., 1992). Finally, humblebragging refers to making a seemingly modest statement that is meant to draw attention to one's admirable or impressive qualities or achievements (Humblebrag, 2017; Wittels, 2012). In Appendix C, we provide examples of CEOs bragging, being modest, and humblebragging on Twitter.

We predict that investors' reaction to a CEO's communication style depends on investors' expectations of the disclosure medium. Expectations reflect the anticipated behavior of another person in a particular situation (Burgoon, 1978, 1993). The disclosure medium sets investors' expectations, which are then either confirmed or disconfirmed with each disclosure. Expectancy violations theory suggests that if individuals expect one thing but see something different, it violates their expectations and draws their attention. This increased attention causes them to spend more time processing and evaluating the violation. Based on their evaluation, they interpret the violation in a positive or negative manner, which affects their perception of the violator's credibility and influences subsequent judgments and decisions (Burgoon, 1978, 1993; Clor-Proell, 2009; Griffin & McClish, 2011). In our context, if investors expect a particular communication style from a CEO, but the CEO surprises them and uses a different communication style, they react by questioning the CEO's behavior and credibility. If the CEO is presenting positive information, we expect questioning the CEO's behavior and credibility will mute investors' reaction to the positive information, thereby lowering their perception of the company's investment potential.

We expect that investors' expectations of social media differ from their expectations of traditional forms of CEO communication, like conference calls. Conference calls are a venue where the CEO typically highlights the successes and positive qualities of their firm (Bowen, Davis, & Matsumoto, 2002; Kimbrough & Wang, 2013; Larcker & Zakolyukina, 2012; NIRI, 2013; Rehm, 2013). Therefore, investors likely expect some degree of CEO bragging about their accomplishments during a conference call. On the other hand, social media is a venue for CEOs to interact with investors and other stakeholders on a more personal level (Elliott, Grant, & Hobson, 2018a; Elliott, Grant, & Hodge, 2018b; Kaplan & Haenlein, 2010). Psychology research has shown that in such social settings, modesty tends to strengthen relationships, while bragging tends to weaken them (Davis et al., 2013; Forsyth, Berger, & Mitchell, 1981; Kelley, 1987).

These arguments lead to an interaction between disclosure medium and a CEO's communication style. In our experimental setting, they imply that investors who read a CEO's comments about his firm's positive performance made during a conference call will view the CEO and his message less positively when the CEO violates expectations by being modest rather than bragging. In contrast, they predict that investors who read a CEO's comments about his firm's positive performance on Twitter will perceive the CEO and his message less positively when the CEO violates expectations by bragging rather than being modest. Underlying both of the above predictions is our expectation that questioning the CEO's behavior will affect assessments of the CEO's credibility, which in turn, will influence investment decisions. Finally, we expect that when investors read a CEO's comments about his firm's positive performance and the CEO humblebrags, investors will view the CEO least positively regardless of the disclosure medium; no one likes humblebragging regardless of how it is disclosed.

We test our hypotheses using a 2 × 2 + 2 between-subjects experiment. Our independent variables are communication style (brag vs. modest + humblebrag) and disclosure medium (conference call vs. Twitter). In our study, graduate and senior undergraduate accounting students assume the role of prospective investors evaluating a potential investment. Participants first learn about a firm and its CEO, and then view a press release containing the firm's 3rd Quarter financial results. The press release contains financial metrics and a narrative section that indicate that the firm performed better than the same quarter last year, with increases in revenue, earnings, and cash flow from operations. In the narrative section, the increases in revenue, earnings, and cash flows are each attributed to both an internal and external event.

After making a preliminary investment assessment, participants read comments from the CEO about the firm's performance. Embedded in the comments is our communication style manipulation. The CEO's comments reflect bragging or modesty. We run two additional conditions where the CEO's comments reflect humblebragging (making a seemingly modest statement that is meant to draw attention to one's admirable or impressive qualities or achievements). We hold information constant across conditions; the CEO's comments only refer to attributes already stated in the press release. We manipulate our second independent variable, disclosure medium, by providing the CEO's comments via a conference call transcript or via Twitter. After viewing the CEO's comments, participants make a final investment judgment, answer questions about the CEO's credibility, and complete the post-experiment questionnaire.

Results support our predictions, and reflect that a CEO's communication style interacts with disclosure medium to influence investor perceptions and investment judgments. Specifically, we find that participants who read a CEO's comments about his firm's positive performance made on a conference call (on Twitter) are *less* likely to invest in the firm when the CEO is modest (brags) compared to when the CEO brags (is modest). We also find, consistent with theory, that perceived CEO credibility mediates the relationship between our dependent variable (willingness to invest) and the interaction between communication style and disclosure medium. Finally, regardless of the disclosure medium of the CEO's comments, we find that participants are least willing to invest in the firm when the CEO humblebrags.

Our study makes several contributions. First, we contribute to the emerging literature on social media and disclosure (Blankespoor, Miller, & White, 2014; Miller & Skinner, 2015). Recent research has examined benefits of CEOs communicating with investors through Twitter compared to traditional mediums while holding constant the style of the communication (Elliott et al., 2018b). We contribute by providing evidence of the importance of matching the CEO's communication style with disclosure medium expectations. We find that the communication style that is effective when using traditional disclosure mediums (like conference calls) may backfire when using new disclosure mediums (like Twitter), causing investors to react negatively. We use an experiment because it allows us to hold constant factors that have been previously shown to affect investor perceptions (e.g., macroeconomic conditions, firm characteristics, management's incentives, etc.), and allows us to isolate the effects of a CEO's communication style. While it is possible to examine CEO sentiment (Chen, Hwang, & Liu, 2017; Davis, Ge, Matsumoto, & Zhang, 2015), measuring the various communication styles using archival methodologies would be difficult. An experiment allows us to test for these effects using a counterfactual, which will likely never be cleanly observable in an actual business setting as CEOs self-select into communication styles. Additionally, an experiment allows us to assess investor perceptions of CEO credibility, an important determinant of investor judgments.

Second, we contribute to the literature investigating how style

[1] Hales et al. (2011) examine the construct of vivid language, which they define as "(a) emotionally interesting, (b) concrete and imagery-provoking, and (c) proximate in a sensory, temporal, or spatial way." To operationalize our bragging manipulation, we examine the construct of intensifying language which includes emotionally-laden words as well as specific graphic language. Intensifying language is a style feature related to word choice, and thus is a narrower construct than vivid language (Hosman, 2002). For example, prior research also considers vivid language to be that which is communicated face-to-face as opposed to in print (Herr, Kardes, & Kim, 1991), or has specific grammatical structures such as active voice and present tense (Burns, Biswas, & Babin, 1993). We operationalize our bragging manipulation using intensifying language to hold these other factors constant across conditions and only vary the communication style.

features of disclosures influence investor judgments (Libby & Emett, 2014). We examine the effects of two communication styles—bragging and modesty—that are present to varying degrees in almost every firm or CEO disclosure that contains more than mere facts. We extend prior literature examining components of these communication styles, namely internal versus external attribution (Aerts, 2005; Baginski, Hassell, & Kimbrough, 2004; Barton & Mercer, 2005; Chen, Han, & Tan, 2016; Elliott, Hodge, & Sedor, 2011; Osma & Guillamón-Saorín, 2011; Solomon, Solomon, Joseph, & Norton, 2013). Specifically, we provide evidence that investors' reaction to these communication styles depends on disclosure medium expectations. Our results suggest that disclosure medium expectations could be an important moderator of other style features documented in prior research. These findings are especially important as firms and investors increasingly use social media (Brandfog, 2016; Weber Shandwick, 2017). Additionally, we introduce humblebragging to the accounting literature, which is an increasingly popular communication style used on social media. We extend prior literature examining humblebragging (i.e., Sezer et al., 2018) by providing evidence that regardless of the disclosure medium, investors perceive a CEO who humblebrags negatively.

Finally, our study has practical implications for managers as they increasingly disclose information via social media outlets like Twitter. Many firms utilize Twitter as a complementary medium to traditional channels. For example, by live tweeting quotes from conference calls (Joyce, 2013). This study informs managers that the most effective communication style for conference calls is not necessarily the most effective communication style for Twitter.

2. Background and hypothesis development

In 2013, the SEC announced that firms could use social media platforms like Twitter to disclose material information in compliance with Regulation Fair Disclosure. Since then, CEOs have increased their use of social media to communicate with stakeholders (Barnes & Lescault, 2011; Barnes & Griswold, 2016; SEC, 2013). Fifty percent of CEOs of the top Fortune 500 companies communicate online via social media, up from 28% in 2010 (Weber Shandwick, 2017). Internally, firms view the increase in CEO presence on social media as a benefit and an opportunity. A survey of Fortune 1000 executives found that 93% of the respondents believe that it is vital for CEOs to participate actively on social media to attract and build connections with investors (Brandfog, 2016). On average, executives attribute 45% of the firm's reputation to that of the CEO's reputation on social media. Consistent with this belief, research has shown that tweets by CEOs elicit a market reaction and increase investor and customer base (Chen et al., 2017). Under the social media umbrella, Twitter is a social networking service that allows users to send and read short 140-character messages called tweets. We focus on Twitter because it is the dominant social media tool for Investor Relations work (NIRI, 2013), and theory suggests that the communication styles that CEOs have successfully used in traditional disclosure mediums (e.g., conference calls) may not be effective on disclosure venues like Twitter.

2.1. Communication style

All accounting disclosures have two components: the content and the style with which the content is communicated (Libby & Emett, 2014). Firms vary both content (Aerts, 2005; Cho, Roberts, & Patten, 2010; Merkl-Davies, Brennan, & McLeay, 2011; Osma & Guillamón-Saorín, 2011; Solomon et al., 2013), and style features, the latter of which is the focus of this paper. Prior research has shown that differences in communication styles such as tone, numerical intensity, and vividness of language, influence investor reactions to a disclosure (Davis & Tama-Sweet, 2011; Davis et al., 2015; Demers & Vega, 2014; Hales et al., 2011; Henry, 2008; Li, 2010; Loughran & McDonald, 2016; Rennekamp, 2012; Rogers, Van Buskirk, & Zechman, 2011; Tan et al., 2014). We focus on two widely observed communication styles used to influence how individuals perceive a presenter (Leary & Kowalski, 1990; Marwick & Boyd, 2011; Smith, 2010; Sezer et al., 2018; Tice et al., 1995; Wosinka et al., 1996), and one new style that has grown along with the use of social media: humblebragging (Sezer et al., 2018).[2]

CEOs exhibit bragging, modesty, and humblebragging across many disclosure mediums. Bragging refers to attributing success to internal factors such as one's own ability or effort (Wosinka et al., 1996; Zuckerman, 1979) and is typically accompanied by intensifying language. Intensifying language includes emotionally-laden words as well as specific graphic language, such as 'innovative,' 'state-of-the-art,' 'gigantic,' and 'slash' (Hosman, 2002).[3] Conversely, modesty refers to attributing success to external factors such as luck or assistance without the use of intensifying language (Carlston & Shovar, 1983; Miller et al., 1992; Tice et al., 1995).

Our paper extends prior literature that mainly focuses on the internal versus external attribution dimensions of these communication styles. Archival research on internal versus external attribution (e.g., Aerts, 2005; Baginski et al., 2004; Osma & Guillamón-Saorín, 2011; Solomon et al., 2013) offers no counterfactual, and therefore, no way to compare the difference in investor reaction if the firm had chosen a different communication style. Experimental research on internal versus external attribution has mostly focused on traditional disclosure mediums (Barton & Mercer, 2005; Chen et al., 2016).[4] We compare a traditional disclosure medium (conference calls) to a new disclosure medium (Twitter) and focus on the interaction of communication style and disclosure medium after the CEO describes positive firm performance.

Our paper also addresses a new communication style growing in popularity along with the use of social media venues like Twitter: humblebragging (Sezer et al., 2018). Humblebragging is making a seemingly modest statement that is meant to draw attention to one's admirable or impressive qualities or achievements (Humblebrag, 2017; Wittels, 2012). Humblebragging is distinct from bragging and modesty because while humblebraggers attempt to appear modest, their motive of bragging is transparent, which causes them to be perceived as being dishonest (Sezer et al., 2018). Recent surveys show that CEOs face pressure from firm employees to not only be modest in their communication, but also be active on social media and promote the firm (KRC Research & Weber Shandwick, 2015). Thus, even CEOs have begun to humblebrag on social media (see Appendix C for an example). Humblebragging is overwhelmingly perceived negatively (Alford, 2012; Sezer et al., 2018; Wittels, 2011).

[2] Earlier research examining attribution typically provides participants with only one level of the attribution manipulation, thus also varying information content (e.g., Barton & Mercer, 2005; Elliott et al., 2011). However, recent research examining attribution provides participants with both levels of the attribution manipulation and instead highlights one of the levels, thus holding information constant across conditions (e.g., Chen et al., 2016). Our design is similar to the latter, in that participants in all conditions receive the same press release containing both attributions. Our manipulation varies the attribution the CEO later highlights from the press release. Since information content remains constant across conditions, we refer to this independent variable as the CEO's communication style. This terminology is also consistent with how the communication literature refers to bragging and modesty (e.g., Aune & Kikuchi, 1993; Burgoon, Jones, & Stewart, 1975; Carlston & Shovar, 1983; Miller et al., 1992; Ou et al., 2014; Peterson & Seligman, 2004; Tangney, 2002; and; Tice et al., 1995).

[3] Results of a pilot study with 39 undergraduate students confirm these definitions. We discuss the design and results of the pilot study in greater detail in Section 3.4.

[4] One exception is Elliott et al. (2011), who examine the interaction of disclosure medium (text-based press release or video) and the CEO's attribution (internal or external) for a *negative* event (restatement). They find that video, a richer medium, amplifies investors' reactions to the CEO's attribution for the negative event relative to text. In our setting, we expect and find that investors react to whether their expectations match a CEO's communication choices. We discuss the match between investors' expectations and a CEO's communication choices in Section 4.5.

2.2. Disclosure medium expectations

We expect that how investors react to a CEO's communication style depends on their expectations of the disclosure medium. Expectations reflect the anticipated behavior of another person in a particular situation (Burgoon, 1978, 1993). The disclosure medium sets investors' expectations, which are then either confirmed or disconfirmed with each disclosure. We expect that investors' expectations of a CEO's communication style differ for two prominent disclosure mediums used by CEOs: conference calls and social media outlets, such as Twitter. During a conference call, investors can call in to hear management report quarterly results, as well as discuss projected earnings (Bowen et al., 2002; NIRI, 2013; Rehm, 2013). Conference calls are a venue for the CEO to highlight the firm's strengths and successes and discuss the future outlook of the firm to attract and retain investors (Bowen et al., 2002; NIRI, 2013; Rehm, 2013).[5] During the scripted performance overview part of a conference call, CEOs often attribute positive firm performance to internal factors and use intensifying words (Bowen et al., 2002; Kimbrough & Wang, 2013; Larcker & Zakolyukina, 2012; NIRI, 2013; Rehm, 2013). Therefore, investors likely expect some degree of CEO bragging about their accomplishments during a conference call.

Conversely, social media platforms like Twitter are "forms of electronic communication through which users create online communities to share information, ideas, personal messages, and other content" (Social Media, 2017). In other words, social media platforms like Twitter are fairly unstructured venues for CEOs to interact with investors and other stakeholders on a more personal level (Elliott et al., 2018b, 2018a; Kaplan & Haenlein, 2010). While both conference calls and Twitter provide opportunities for two-way interactions, features of Twitter create stronger perceptions of a social interaction and thus, change expectations of a CEO's communication style.

Conference calls are formally initiated, led, and ended by firm managers. The managers dictate how the call will proceed and when participants (investors and analysts) can ask questions. Questions asked are censored in the sense that the firm reviews questions before selecting which ones to answer. Twitter provides a much more interactive communication environment. The interactions are typically less formal making them more personal, and allow anyone following a firm or its CEO to ask questions quickly and with low effort, whenever they want. These questions are not vetted by the firm before others see them. Additionally, via retweets and likes, investors can easily put pressure on managers to respond to questions asked but not yet answered. This interactive environment creates a greater capacity to facilitate shared meaning, which makes Twitter a more social setting compared to conference calls. Individuals have expectations about what constitutes an appropriate communication style in social settings. When these expectations are met, it strengthens social bonds (Kaplan & Haenlein, 2010; Lengel & Daft, 1989; Murthy, 2013). Psychology research has shown that modesty can promote the strengthening of social bonds, while bragging can weaken such bonds (Davis et al., 2013; Forsyth et al., 1981; Kelley, 1987). The above discussion suggests that investors have expectations about appropriate communication styles in social settings like Twitter, and that these expectations favor modesty over bragging.

[5] Prior research supports our assumption that retail investors use conference calls when making investment decisions. First, in a survey of 20,000 retail investors, 47.9% indicated that they read conference call transcripts (PR Newswire, 2014). Further, prior research suggests that firms care about how retail investors perceive firm communications, such as conference calls (Graham, Harvey, & Rajgopal, 2005) and that retail investors trade based on conference calls. Specifically, archival research finds that open conference calls (i.e., calls open to all investors, including retail) are associated with an increase in the percentage of small trades, while closed conference calls are not (i.e., calls usually limited to analysts and large institutional investors) (Bushee, Matsumoto, & Miller, 2004).

2.3. The interaction of communication style and disclosure medium

We expect that a CEO's communication style and disclosure medium interact to influence how investors perceive the CEO by altering their perceptions of the CEO's credibility. CEO credibility reflects investors' perceptions of the CEO's trustworthiness and competence in financial reporting and managing the firm (Giffin, 1967; Mercer, 2005). Perceived CEO credibility is important for how both analysts and investors evaluate a firm (Demerjian, Lev, & McVay, 2012; Hewitt, Hodge, & Pratt, 2017; Malmendier & Tate, 2005, 2008; Mercer, 2005; Palmrose, Richardson, & Scholz, 2004; Prowse, 1998). When an individual is perceived as more credible, their communications have a greater influence on investment decisions (Hirst, Koonce, & Venkataraman, 2008; Hodge, Hopkins, & Pratt, 2006; Mercer, 2005).

Expectancy violations theory explains how individuals respond to confirmations and violations of expectations. If individuals see communication consistent with their expectations, they will have no reaction (Burgoon & Burgoon, 2001). However, if individuals expect one thing, but see something different, it violates their expectations and catches their attention. This increased attention causes them to spend more time processing and evaluating the violation. Based on their evaluation, they interpret the violation in a positive or negative manner, which affects their perception of the violator's credibility and influences subsequent judgments and decisions (Burgoon, 1978, 1993; Clor-Proell, 2009; Griffin & McClish, 2011). Interpretations typically lead to strong negative reactions if the violator communicates in a way that makes the violator's ulterior motive transparent (Jones & Pittman, 1982; Sezer et al., 2018). This is typically the case with humblebraggers, who are attempting to be modest while being perceived as bragging (Sezer et al., 2018).

In our setting, investors expect CEOs to brag on conference calls, so a CEO who is modest on a conference call violates investors' expectations. This violation catches investors' attention, leading them to question the CEO's behavior and credibility: why is the CEO not taking credit for his company's performance in a setting where investors expect a CEO to take credit for such performance? If the CEO is presenting positive information, we expect questioning the CEO's behavior and credibility to mute investors' reactions to the positive information, thereby lowering their perception of the company's investment potential. We expect a similar reaction if a CEO violates investors' expectations on Twitter. Twitter is a social medium that creates social bonds. In such settings people expect modesty, and dislike bragging (Davis et al., 2013; Forsyth et al., 1981; Kelley, 1987). When a CEO violates expectations by bragging on Twitter, it draws investors' attention, causing them to question the CEO's behavior and credibility. Like before, if the CEO is presenting positive information, we expect questioning the CEO's behavior and credibility to mute investors' reactions to the positive information, thereby lowering their perception of the company's investment potential. Finally, we expect humblebragging, regardless of disclosure medium, to be viewed negatively. Humblebragging allows investors to see through the CEO's ulterior motive, which is to brag about performance while attempting to appear modest.

In summary, we expect that investors who read a CEO's comments about his firm's positive performance made during a conference call will view the CEO and his message less positively when the CEO violates expectations by being modest rather than bragging. Conversely, we expect that investors who read a CEO's comments about his firm's positive performance on Twitter will view the CEO and his message less positively when the CEO violates expectations by bragging rather than being modest.[6] Underlying both of the above predictions is our

[6] Ex ante we are uncertain if investors will view the CEO more negatively when the CEO is modest on a conference call compared to Twitter or if investors' judgments will not differ because modesty is highly valued in general (Ben-Ze'ew, 1993; Ou et al., 2014; Wosinka et al., 1996). In other words, we do not make a directional simple effect prediction about the effect of disclosure medium given the CEO is modest. Therefore, our interaction prediction could take the form of either an ordinal or disordinal pattern.

expectation that questioning the CEO's behavior will affect assessments of the CEO's credibility, which in turn, will influence investment decisions. Finally, we expect that when investors read a CEO's comments about his firm's positive performance, and the CEO humblebrags, investors will view the CEO least positively regardless of the disclosure medium. Formally, our hypotheses are:

H1. *When the disclosure medium of a CEO's comments is a conference call (Twitter), investors will be more (less) willing to invest in the firm when the CEO brags compared to when the CEO is modest.*

H2. *Perceived CEO credibility will mediate the influence of a CEO's communication style and disclosure medium on investors' willingness to invest in the CEO's firm.*

H3. *Regardless of the disclosure medium of the CEO's comments, investors will be less willing to invest in a firm when the CEO humblebrags compared to when the CEO brags or is modest.*

3. Experimental method

To test our predictions, we conduct an experiment with a 2 × 2 + 2 between-subjects design, with communication style and disclosure medium as the manipulated independent variables.[7] We manipulate communication style by varying whether the CEO's comments reflect bragging or modesty. We manipulate disclosure medium by presenting the CEO's comments via a conference call transcript or via Twitter. We run two additional conditions where the CEO's comments reflect humblebragging as reflected in a conference call transcript or on Twitter.

3.1. Participants

Participants are 145 graduate and senior undergraduate accounting students from a large, public university.[8] The participants have taken on average 11.1 accounting courses and 3.0 finance courses. Overall, 33.3 percent of participants have purchased common stock or debt securities, while 78.8 percent plan to do so in the next five years. We choose student subjects for several reasons. First, we match participants' knowledge to the task without using more sophisticated subjects than necessary (Libby, Bloomfield, & Nelson, 2002). Second, the students we recruit have reasonable familiarity with social media and the information disclosed in conference calls. Participants complete the study via an online instrument within the lab.

3.2. Task and procedures

Participants arrive at a computer lab session, sit at a computer, and are randomly assigned to a condition by the computer software. We instruct participants to assume the role of investors who own a portfolio of stocks. As part of the annual review of their portfolio holdings, they are considering investing in the semi-conductor industry to maintain a diversified portfolio. The firm they are considering is HSG Inc., a fictitious firm in the semi-conductor industry. After reading instructions, participants view background information about the firm, its industry, and its CEO, Roger Smith. Next, they view a quarterly earnings press release containing financial metrics and a narrative section. The financial metrics show that revenue, earnings, and cash flow from operations increased from the same quarter last year. The narrative section contains bullet points that repeat the growth in revenue, earnings, and cash flows and attribute each increase to both an internal and an external event. For example, the cash flow from operations bullet point states: "Cash Flow from Operations were $1845 for the quarter, up 3.13% from same quarter last year. Strong macroeconomic conditions and improved receivable collection practices contributed to this increase." The firm attributes its increase in cash flow from operations to improved receivable collection practices (internal attribution), and also to strong macroeconomic conditions (external attribution). Appendix A contains images of the press release.

3.3. Preliminary investment assessment

After participants view the press release, they make a preliminary willingness to invest assessment, which we measure using two questions. First, we ask: "How attractive is an investment in HSG stock as part of your diversified portfolio?" Participants respond on a 101-point scale with endpoints 0 ("Very Unattractive") and 100 ("Very Attractive"). Second, we ask: "How likely are you to invest in HSG stock as part of your diversified portfolio?" Participants respond on a 101-point scale with endpoints 0 ("Very Unlikely") and 100 ("Very Likely").

3.4. Communication style manipulation

After indicating their preliminary willingness to invest, participants view a screen that states they will next view the CEO's comments regarding HSG's Q3 2016 results (i.e., comments about the press release they previously viewed). Participants then click to the next screen, which contains our communication style manipulation. Our manipulation varies whether the CEO brags (internal attribution and presence of intensifying language) or is modest (external attribution and absence of intensifying language) about the firm's positive performance.

In both our brag and modesty conditions, the CEO makes three statements, which discuss the increases in revenue, earnings, and cash flow from operations. Recall that the press release attributes increases in these three items to both internal and external factors. Therefore, we hold information content constant across all conditions as the CEO repeats one of the two attributions from the press release. In brag conditions, the CEO attributes the growth in revenue, earnings, and cash flows to the internal attributions discussed in the press release. Additionally, the CEO uses intensifying language when discussing results. For example, one of the statements in the CEO's comments is "our innovative new collection practices caused a gigantic surge in cash flow from operations." In this statement, the CEO attributes the increase in revenue to an internal factor and uses intensifying language ("innovative," "gigantic," and "surge").

In modesty conditions, the CEO attributes the growth in revenue, earnings, and cash flows to the external attributions discussed in the press release and does not use intensifying language. For example, one of the statements in the CEO's comments is "strong macroeconomic conditions caused an increase in cash flows from operations." In this statement, the CEO attributes the increase in revenue to an external factor and does not use intensifying language.

We include two additional conditions where the CEO humblebrags. In humblebrag conditions, the CEO makes seemingly modest statements regarding the increases in revenue, earnings, and cash flows, but then draws attention to the firm's impressive qualities. For example, one of the statements in the CEO's comments is "strong macroeconomic conditions caused an increase in cash flows from operations. Apparently, we will have to wait for a down market to see the true effect of our innovative collection practices." The CEO initially attributes the increase in revenue to an external factor (identical to the modesty conditions), but then draws attention to an internal factor and uses intensifying language ("innovative new collection practices"). Appendix B contains images of our manipulations.

[7] We describe the experiment as a 2 × 2 + 2 between-subjects design (as opposed to a 3 × 2) because bragging and modesty are considered endpoints of the same communication style construct continuum, while humblebragging is considered a separate communication style construct (Sezer et al., 2018).

[8] Our participants are a combination of graduate and senior undergraduate accounting students (approximately 50% each). We use two pools of participants to ensure a large enough sample size, and because theoretically we do not expect program type to interact with our manipulations. Results are inferentially similar using just graduate students or just senior undergraduate students in our tests.

Our brag manipulation jointly manipulates internal attribution and intensifying language. We chose to do so based on theory and the results of a pilot study. The pilot study consisted of 39 undergraduate students. We ask all pilot participants the following three questions about bragging [three questions about modesty] for a total of six questions: 1) "When a person is bragging [making a modest statement], they attribute their success to internal factors (i.e., their own ability or effort)," 2) "When a person is bragging [making a modest statement], they attribute their success to external factors (i.e., luck or others' assistance)," and 3) "When a person is bragging [making a modest statement], they likely use intensifying adjectives, verbs and adverbs. For example, words such as important, large, extensive, terrific, amazing, special, unique, excellent, demolished, crushed, destroyed, highly, strongly, ridiculously, deeply, considerably, definitely, extremely." We randomize whether participants respond to the three bragging or modesty questions first. Participants respond to all questions on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree").

Pilot results support our expectations. First, participants more strongly agree that a person attributes their success to internal factors when they brag than when they are modest ($87.26 > 38.31$; $t_{(38)} = 13.40$, $p < 0.01$, one-tailed). Second, participants more strongly agree that a person attributes their success to external factors when they are modest than when they brag ($73.69 > 18.49$; $t_{(38)} = 15.60$, $p < 0.01$, one-tailed). Finally, participants more strongly agree that a person uses intensifying language when they brag than when they are modest ($78.59 > 32.64$; $t_{(38)} = 11.50$, $p < 0.01$, one-tailed). These results provide support for our choice to use a joint manipulation for bragging in our primary experiment. We realize, however, that our manipulation is a compound manipulation, which makes clearly interpreting our results difficult. To help clarify what is driving any observed effects, we also randomly assign participants to two additional internal attribution conditions which do not include intensifying language. These additional cells allow us to disentangle the influence of internal attribution and intensifying language, which we do in Sections 4.6 and 4.7.

3.5. Disclosure medium manipulation

We manipulate disclosure medium by presenting the CEO's comments via a conference call transcript or Twitter. We operationalize a conference call transcript rather than an audio-based conference call to avoid manipulating multiple conceptual variables. Our choice is realistic as a survey of 20,000 retail investors reflects that 47.9 percent use conference call transcripts (PR Newswire, 2014). Participants in Twitter conditions receive the CEO's comments in the form of three separate tweets presented on the same screen. Our Twitter manipulation purposefully abstracts away from many features of Twitter, like the Twitter logo and bird, information about retweets, likes, etc., and the ability to respond any of the tweets (i.e., mundane realism). Consistent with prior research, we expect that seeing a Twitter-like interface evokes *perceptions* of the social interaction features of the medium, regardless of whether the features actually appear or not (Elliott et al., 2018b).[9] In all conditions, participants view the CEO's picture and the time of the communication to strengthen perceptions that the comments came from the CEO.

[9] We confirm in out-of-sample data that our abstract Twitter manipulation does evoke higher perceptions of social interaction with the CEO than our conference call manipulation. Specifically, 117 students respond to the following three questions adapted from scales developed to capture social interaction (e.g., Leiner & Quiring, 2008; Liu, 2003; O'Brien & Toms, 2010; Sohn & Choi, 2014): 1) "I felt that I could have back-and-forth communication with Roger Smith," 2) "I felt that communicating with Roger Smith would require very little effort," and 3) "I felt that I could communicate with Roger Smith very quickly." Participants respond on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree"). We conduct a one-way ANOVA with average interaction (Cronbach's Alpha = 0.84) as the dependent variable and disclosure medium as the independent variable. Consistent with our expectations, results (untabulated) reveal a significant effect of disclosure medium, suggesting participants perceived higher interaction with the CEO in Twitter compared to conference call conditions ($50.04 > 44.95$; $F_{(1,115)} = 2.02$, $p = 0.08$, one-tailed).

3.6. Final investment assessment

After viewing the CEO's comments, participants make their final willingness to invest assessment by responding to the same two questions they answered when making their preliminary willingness to invest assessment. We remind participants that their answers should reflect their final assessment and not how they would adjust their preliminary assessment.

3.7. Credibility assessment and other post-experimental questions

After making their final investment judgments, participants answer two questions about their perceptions of the CEO's credibility: 1) "I believe that CEO Roger Smith is trustworthy" and 2) "I believe that CEO Roger Smith is competent." Participants respond to both questions on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree"). Participants then answer manipulation-check questions, and respond to questions about their use of social media, as well as provide demographic information.

4. Results

4.1. Manipulation checks

To assess the effectiveness of the communication style manipulation (i.e., bragging or modesty), we ask participants two questions: 1) "I believe that CEO Roger Smith was bragging when he communicated about HSG's Q3 2016 results" and 2) "I believe that CEO Roger Smith was being modest when he communicated about HSG's Q3 2016 results." Participants respond to both questions on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree"). Participants in bragging conditions believe the CEO was bragging more than participants in modesty conditions ($59.90 > 40.60$; $t_{(96)} = 5.30$, $p < 0.01$, one-tailed). Conversely, participants in modesty conditions believe the CEO was more modest than participants in bragging conditions ($56.46 > 43.10$; $t_{(96)} = 3.62$, $p < 0.01$, one-tailed).

To assess the effectiveness of the humblebragging manipulation, we ask participants "I believe that CEO Roger Smith was humblebragging (trying to appear modest while actually bragging) when he communicated about HSG's Q3 2016 results." Participants respond on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree"). Participants in humblebragging conditions believe the CEO was humblebragging more than participants in modesty conditions ($62.07 > 47.41$; $t_{(49)} = 2.50$, $p = 0.01$, one-tailed), but to a similar extent as participants in bragging conditions ($62.07 \approx 57.71$; $t_{(55)} = 0.60$, $p = 0.28$, one-tailed).

Given the mixed results of our humblebragging manipulation check, we decided to ask a second communication style manipulation check out-of-sample using 117 graduate and senior undergraduate accounting student participants. Participants are provided a) the definitions of bragging, modesty, and humblebragging and b) the three communication style manipulations from our primary experiment. Participants are then asked to assign one definition to each of the communication style manipulations. Consistent with our expectations, most participants match the bragging manipulation and definition (67% of participants; $\chi^2_{(2)} = 67.85$, $p < 0.01$), the modesty manipulation and definition (82% of participants; $\chi^2_{(2)} = 127.13$, $p < 0.01$), and the humblebragging manipulation and definition (57% of participants; $\chi^2_{(2)} = 35.28$, $p < 0.01$). These out-of-sample results provide some additional comfort that our communication style manipulation captures our intended construct.

To assess the effectiveness of the disclosure medium manipulation, we ask participants: "How did you receive the information from CEO Roger Smith?" with answer choices that show images of the blurred-out tweets and blurred-out conference call transcript along with headings that state "Tweets" and "Conference call," respectively. Ninety-six

percent of participants correctly answer this question. It appears our manipulations were successful. Results are inferentially similar if we exclude participants who fail this manipulation check question.

4.2. Test of hypothesis 1 – willingness to invest

H1 predicts that when the disclosure medium of a CEO's comments is a conference call (Twitter), investors will be more (less) willing to invest in the firm when the CEO brags compared to when the CEO is modest. We measure preliminary and final willingness to invest by averaging participants' ratings of 1) the attractiveness of an investment in the firm's stock and 2) the likelihood that they would invest in the firm's stock. A Cronbach's alpha of 0.93 (0.94) for our two final (preliminary) ratings confirms that our investment-related questions are capturing the same underlying construct. We calculate the change in participants' willingness to invest by subtracting their final willingness to invest rating from their preliminary willingness to invest rating.[10] Negative values indicate a decrease in willingness to invest after viewing the CEO's comments. Table 1, Panel A reports cell sizes, means, and standard deviations for the change in willingness to invest. Fig. 1 displays the pattern of cell means for our dependent variable.

Table 1, Panel B presents a two-way ANOVA with change in willingness to invest as the dependent variable. Results reveal a significant communication style by disclosure medium interaction ($p < 0.01$, one-tailed).[11] We report the follow-up simple effects tests in Table 1, Panel C. Results show that when the disclosure medium of the CEO's comments is a conference call, participants are *less* willing to invest when the CEO is modest compared to when the CEO brags ($p = 0.09$, one-tailed). However, when the disclosure medium is Twitter, participants are *less* willing to invest when the CEO brags compared to when the CEO is modest ($p < 0.01$, one-tailed). These results are consistent with our hypothesis that investors will be less willing to invest when a CEO uses a communication style that does not align with the expectations of the disclosure medium.[12, 13, 14]

Table 1
How communication style and disclosure medium affect willingness to invest—tests of H1 and H3.

Panel A: Change in Willingness to Invest, Mean (Standard Deviation), n = 145[a]

Communication Style	Disclosure Medium	
	Conference Call	Twitter
Brag	1.15	−6.92
	(8.15)	(13.70)
	n = 24	n = 26
Modest	−2.85	−0.38
	(7.56)	(9.86)
	n = 24	n = 24
Humblebrag	−14.02	−12.37
	(18.75)	(13.99)
	n = 24	n = 23

Panel B: Two-Way ANOVA Model of Change in Willingness to Invest (Excluding Humblebrag Conditions)—H1[b]

Source of Variation	Sum of Squares	df	Mean Square	F	Sig.
Communication Style	39.72	1	39.72	0.38	0.54
Disclosure Medium	191.15	1	191.15	1.84	0.18
Communication Style x Disclosure Medium	680.66	1	680.66	6.55	0.01[e]
Error	9768.45	94	103.92		

Panel C: Follow-Up Simple Effects Tests (Excluding Humblebrag Conditions)—H1[c]

Source of Variation	t	Sig.
Effect of Communication Style given Conference Call	1.36	0.09[e]
Effect of Communication Style given Twitter	2.27	0.01[e]
Effect of Disclosure Medium given Brag	2.80	0.01[e]
Effect of Disclosure Medium given Modest	0.84	0.40

Panel D: Planned Comparisons of Humblebrag Conditions—H3[d]

Comparison	t	Sig.
Conference Call/Humblebrag < Conference Call/Brag	3.63	0.01[e]
Conference Call/Humblebrag < Conference Call/Modest	2.71	0.01[e]
Twitter/Humblebrag < Twitter/Brag	1.38	0.09[e]
Twitter/Humblebrag < Twitter/Modest	3.41	0.01[e]

[a] Table 1 presents tests of H1. Panel A presents descriptive statistics. The dependent variable is change in willingness to invest. Participants rated 1) the attractiveness of an investment in the firm's stock and 2) the likelihood that they would invest in the firm's stock. The willingness to invest variable is simply an average of these measures. Our dependent variable, change in willingness to invest, is final willingness to invest minus preliminary willingness to invest. Fig. 1 provides an illustration of these results. In the primary experiment, we manipulate communication style by providing comments from the CEO about the firm's performance that reflect bragging or modesty. We run two additional conditions where the CEO's comments reflect humblebragging (trying to appear modest while actually bragging). We manipulate disclosure medium by providing the CEO's comments via a conference call transcript or via Twitter.

[b] Panel B presents an ANOVA for the four primary conditions.

[c] Panel C presents follow-up simple effects tests for the four primary conditions.

[d] Panel D presents planned comparisons for the humblebrag conditions.

[e] p-values are one-tailed equivalents for directional predictions.

For completeness, we also report the simple effects of disclosure medium when the CEO's communication style is bragging or modesty. As expected, participants are *less* willing to invest when the CEO brags

[10] Consistent with the firm disclosing positive performance, we find that participants' preliminary and final willingness to invest judgments in all four conditions are significantly greater than the midpoint (all p-values < 0.01). As expected, we find no differences in participants' preliminary willingness to invest across conditions ($p = 0.40$, two-tailed). Results are inferentially similar if we use either of the individual investment questions rather than the combined measure. Results are also robust to: 1) running an ANOVA with the final willingness to invest as the dependent variable and preliminary willingness to invest as a covariate and 2) using final willingness to invest as the dependent variable without including preliminary willingness to invest as a covariate.

[11] Participants' willingness to invest judgments are not normally distributed in three conditions and Levene's test for equality of variances indicates the assumption of homogeneity of variances is violated. To correct these issues and assess the robustness of our results, we apply both logarithmic and inverse transformations to willingness to invest judgments in all four conditions. Results using these transformed values are inferentially similar to results using the untransformed values. We report results of the untransformed values for H1 for ease of interpretation.

[12] Results for all hypotheses are robust to including participants' assessments of firm risk and management's forthcomingness, gender, first language, information acquisition, Twitter use, experience with financial statements and investing as controls.

[13] We also test the change in participants' willingness to invest within each condition as additional support for our theory. Specifically, we predict that investors initial positive reactions will decrease when a CEO's communication style does not match their disclosure medium expectations. We find that participants significantly decrease their final willingness to invest from their preliminary willingness to invest only in the conditions where the CEO violates expectations of the disclosure medium. Specifically, participants significantly decrease their willingness to invest when the CEO brags on Twitter ($-6.92 < 0.00$; $t_{(25)} = 2.58$, $p = 0.01$, one-tailed) and when the CEO is modest on an earnings conference call ($-2.85 < 0.00$; $t_{(23)} = 1.85$, $p = 0.04$, one-tailed). However, participants do not significantly change their willingness to invest when the CEO's communication style matches their expectations of the disclosure medium, specifically when the CEO is modest on Twitter ($-0.38 \approx 0.00$; $t_{(23)} = 0.19$, $p = 0.85$, two-tailed) or when the CEO brags on an earnings conference call ($1.15 \approx 0.00$; $t_{(23)} = 0.69$, $p = 0.50$, two-tailed).

[14] According to expectancy violations theory, a violation of expectations attracts attention, causing an individual to spend more time processing and evaluating the violation (Burgoon & Burgoon, 2001). Consistent with theory, results (untabulated) show that

(footnote continued)
participants spend significantly more time examining the CEO's comments and responding to the final willingness to invest and credibility questions when the CEO violates their expectations. Specifically, participants spend significantly more time when the CEO is modest on a conference call or brags on Twitter than when the CEO brags on a conference call or is modest on Twitter ($53.82 > 46.35$; $t_{(95)} = 2.25$, $p = 0.01$, one-tailed).

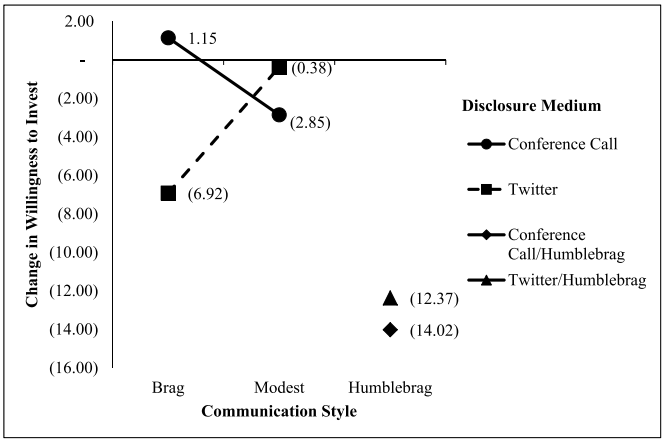


Fig. 1. Primary Experiment—The Effect of Communication Style and Disclosure Medium on Change in Participants' Willingness to Invest.
This figure graphically depicts our observed mean values for the change in participants' willingness to invest in the primary experiment. Negative (positive) values indicate a decrease (increase) in willingness to invest after viewing the CEO's comments, which contain our manipulations. Table 1 presents descriptive statistics for the change in participants' willingness to invest.

on Twitter compared to a conference call ($p < 0.01$, one-tailed).[15] Finally, participants' willingness to invest judgments are insensitive to disclosure medium when the CEO is modest ($p = 0.40$, two-tailed). Finding no difference in participants' willingness to invest when the CEO is modest on a conference call versus Twitter is consistent with modesty being highly valued in general (Ben-Ze'ew, 1993; Ou et al., 2014; Wosinka et al., 1996).

4.3. Test of hypothesis 2 – credibility

H2 predicts that perceived CEO credibility will mediate the influence of a CEO's communication style and disclosure medium on investors' willingness to invest. Table 2, Panel A presents the two questions we use to measure credibility. Table 2, Panel B presents descriptive statistics for both credibility questions and the combined credibility measure. The path model we use to test H2 includes only the four primary conditions (brag and modest; conference call and Twitter), and is presented in Fig. 2, Panel A.

Fig. 2, Panel B presents results consistent with H2.[16] We observe a significant relation between the interaction of our two manipulated independent variables, communication style and disclosure medium, on perceived CEO credibility ($p = 0.04$, one-tailed). We find a significant positive effect of perceived CEO credibility on the change in willingness to invest ($p = 0.01$, one-tailed). Finally, the link between the interaction of our two manipulated independent variables, communication style and disclosure medium, and change in willingness to invest is no longer significant ($p = 0.54$, two-tailed) when we include credibility in the model.

We also use a bootstrap analysis to confirm that perceived CEO credibility mediates the influence of a CEO's communication style and disclosure medium on participants' willingness to invest (Cheung & Lau, 2008). The bootstrapping method generates an empirical sampling distribution of the indirect effect and generates a confidence interval for its value. Drawing a bootstrap sample of $N = 1{,}000$, we find that the mean indirect effect on change in willingness to invest is positive (mean = 0.08, untabulated), with a 90% confidence interval excluding zero (0.006–0.208, untabulated). Our path model and bootstrap analysis both indicate that perceived CEO credibility mediates the influence of communication style and disclosure medium on investment judgments.[17]

Table 2
Descriptive statistics—credibility.

Panel A: Questions Used to Measure Credibility[a]			
1) I believe that CEO Roger Smith is trustworthy. 2) I believe that CEO Roger Smith is competent.			
Panel B: Measures of Credibility, Mean (Standard Deviation), n = 145[b]			
Experimental Condition (Communication Style x Disclosure Medium)	Question		
	1	2	Average Credibility
Brag/Conference Call (n = 24)	61.92 (16.33)	68.17 (15.05)	65.04 (15.05)
Modest/Conference Call (n = 24)	65.88 (12.81)	71.42 (14.11)	68.65 (11.47)
Brag/Twitter (n = 26)	54.85 (17.33)	59.46 (14.57)	57.15 (12.04)
Modest/Twitter (n = 24)	66.38 (14.55)	71.75 (15.74)	69.06 (13.64)
Humblebrag/Conference Call (n = 24)	53.00 (22.30)	57.13 (22.83)	55.06 (21.39)
Humblebrag/Twitter (n = 23)	56.22 (17.01)	54.48 (19.87)	55.35 (16.18)

[a] Table 2 reports descriptive statistics for participants' responses to our 2 questions used to measure perceived CEO credibility by condition. Panel A lists the 2 questions we use to measure perceived CEO credibility. We create a single measure of credibility by averaging these two questions (Cronbach's alpha = 0.80). In the primary experiment, we manipulate communication style by providing comments from the CEO about the firm's performance that reflect bragging or modesty. We run two additional conditions where the CEO's comments reflect humblebragging (trying to appear modest while actually bragging). We manipulate disclosure medium by providing the CEO's comments via a conference call transcript or via Twitter.

[b] Panel B presents means and standard deviations for the 2 questions by condition as well as the average. Participants respond to both questions on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree").

4.4. Test of hypothesis 3 – humblebragging

H3 predicts that regardless of the disclosure medium of the CEO's comments, investors will be less willing to invest when the CEO

[15] An alternative explanation for this result is that participants react negatively to the CEO bragging on Twitter because they perceive three tweets as three instances of bragging instead of one, as in the conference call condition. We rule out this alternative explanation by comparing responses across conditions to our bragging manipulation check question (discussed in section 4.1). We find that participants perceive that the CEO is bragging to a similar extent when comments are provided on Twitter compared to a conference call (57.67 ≈ 61.96; $t_{(48)} = 0.80$, $p = 0.43$ two-tailed).

[16] The fit of the model is good, with a comparative fit index (CFI) of 0.98, minimum fit $\chi^2_{(3\ df)} = 6.06$ ($p = 0.11$), minimum discrepancy divided by degrees of freedom ($\chi 2/df$) of 2.02, RMSEA of 0.10, and SRMR of 0.03. Models with CFI close to 0.95, minimum fit χ^2 $p > 0.05$, $\chi 2/df < 3.0$, RMSEA < 0.10, and SRMR < 0.08 are considered good fits (Hu & Bentler, 1999; Iacobucci, 2010; Marsh et al., 2004).

[17] Inconsistent with our expectations, the pattern of means for the credibility measure does not mirror the pattern of means for change in willingness to invest. Specifically, investors are less willing to invest in the firm when the CEO is modest compared to when the CEO brags on a conference call, but perceived CEO credibility does not differ when the CEO is modest or brags on a conference call. This pattern could reflect that individuals tend to value modesty in general, so when participants in our experiment were questioning the CEO's behavior, they did not significantly lower their assessments of credibility when the CEO was modest (Ben-Ze'ew, 1993). However, the external attributions inherent in modesty could suggest that the firm's positive performance is less persistent (Chen et al., 2016), leading to lower predictions of future earnings and a lower willingness to invest. This would suggest that credibility is an important mediator, but not in all situations.

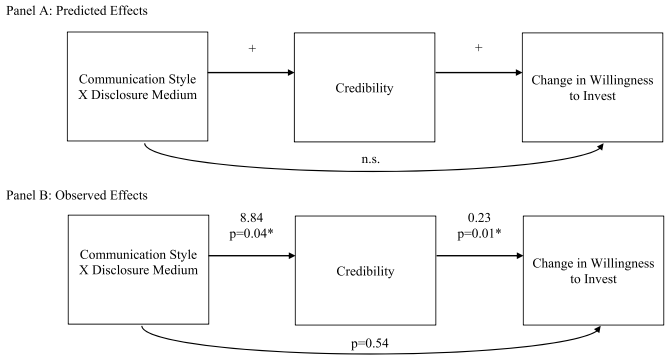


Fig. 2. Path Model—Hypothesis 2 Mediation Analysis.
This figure presents the Hypothesis 2 predicted (panel A) and observed (panel B) pattern that perceived CEO credibility will mediate the influence of communication style and disclosure medium on investors' willingness to invest in a firm. The overall model's comparative fit index (CFI) is 0.98, minimum fit $\chi^2_{(3\ df)} = 6.06$ (p = 0.11), the minimum discrepancy divided by degrees of freedom ($\chi 2/df$) is 2.02, RMSEA is 0.10, and SRMR is 0.03. Models with CFI close to 0.95, minimum fit χ^2 p > 0.05, $\chi 2/df < 3.0$, RMSEA < 0.10, and SRMR < 0.08 are considered good fits (Hu & Bentler, 1999; Iacobucci, 2010; Marsh, Hau, & Wen, 2004). We observe a significant relation between the interaction of our two manipulated independent variables, communication style and disclosure medium, on perceived CEO credibility (p = 0.04, one-tailed). We find a significant positive effect of perceived CEO credibility on the change in willingness to invest (p = 0.01, one-tailed). Finally, the link between the interaction of our two manipulated independent variables, communication style and disclosure medium, and change in willingness to invest is no longer significant (p = 0.54, two-tailed) when we include credibility in the model. Not shown in Fig. 2 but included in our model, we find an insignificant main effect of communication style (p = 0.33, two-tailed) and a significant main effect of disclosure medium (p = 0.03, two-tailed) on credibility.

humblebrags compared to when the CEO brags or is modest. Table 1, Panel A reports cell sizes, means, and standard deviations for the change in willingness to invest. Table 1, Panel D reports the planned comparisons of the humblebrag conditions to the brag and modest conditions. Results show that when the disclosure medium of the CEO's comments is a conference call, participants are *less* willing to invest when the CEO humblebrags compared to when the CEO brags (p < 0.01, one-tailed) or is modest (p < 0.01, one-tailed). Similarly, when the disclosure medium of the CEO's comments is Twitter, participants are *less* willing to invest when the CEO humblebrags compared to when the CEO brags (p = 0.09, one-tailed) or is modest (p < 0.01, one-tailed). Taken together, these results support H3.

Our theory suggests that investors will be least willing to invest when the CEO humblebrags regardless of disclosure medium because they perceive the CEO to be less credible. Consistent with this expectation (results untabulated), we find that participants perceive the CEO to be less credible when the CEO humblebrags compared to when the CEO brags (55.20 < 60.94; $t_{(95)} = 1.71$, p = 0.05, one-tailed) or when the CEO is modest (55.20 < 68.85; $t_{(93)} = 4.18$, p < 0.01, one-tailed).[18, 19]

[18] An alternative explanation for participants' negative reaction to a CEO humblebragging is that they perceived our humblebrag to be incoherent, which we rule out by asking out-of-sample questions. We provide 117 graduate and senior undergraduate accounting students from the same population with the following two statements from our humblebragging manipulation: "An overall increase in consumer demand for wireless products caused an increase in revenue this quarter. Looks like we didn't even need our innovative and groundbreaking marketing campaign," and "Finally, strong macroeconomic conditions caused an increase in cash flows from operations. Apparently, we will have to wait for a down market to see the true effect of our innovative new collection practices." We then ask participants to choose which one of four interpretations most closely matches their interpretation of the sentences, one of which is that the sentences do not make sense and seem incoherent. We find that only 9% ($\chi^2_{(3)} = 90.76$, p < 0.01) and 10% ($\chi^2_{(3)} = 23.41$, p < 0.01) of participants, respectively, interpret the statements as incoherent, providing support that our humblebragging manipulation is interpreted as we intend, and not as incoherent.

[19] For simplicity, we report our mediation analysis excluding our humblebragging conditions. Results are consistent with those reported above (i.e., we observe mediation) if we include the humblebragging conditions in our analysis.

4.5. Further evidence of our theory – disclosure medium expectations

Our theory suggests that investors have disclosure medium expectations, and when a CEO's communication style does not match those expectations, it influences investors' judgments and decisions. We ask disclosure medium expectations questions out-of-sample using 117 graduate and senior undergraduate accounting students. Participants are randomly assigned to either a conference call or Twitter disclosure medium, introduced to a fictional CEO, and view both brag and modesty communication style manipulations from our primary experiment. We then ask: "Which of the following comments would most investors believe that CEO Richard Jones **should** make on **an earnings conference call (Twitter)**?"[20] Participants indicate whether they expect the CEO's comments to reflect either bragging or modesty in that particular disclosure medium. We find that significantly more participants expect the CEO to brag rather than be modest on conference calls (73% > 27%; $\chi^2_{(1)} = 11.36$, p = 0.01). Conversely, significantly more participants expect the CEO to be modest rather than brag on Twitter (66% > 34%; $\chi^2_{(1)} = 6.45$, p = 0.01).

According to expectancy violations theory, beliefs about how an individual *should* communicate drive expectations during new encounters (Burgoon & Burgoon, 2001). As participants in our study have no prior history with the CEO, we chose to ask them about how a CEO *should* communicate as opposed to how a CEO *would* communicate. To validate that framing our question in terms of *should* results in participants experiencing an expectations violation when they learn of the CEO's behavior, we run an out-of-sample experiment using 109 graduate and senior undergraduate accounting students. We randomly assign participants to either a conference call or Twitter disclosure medium, and inform them that the CEO is going to disclose a set of comments, but has not yet decided which set to disclose. One set of comments reflects our brag manipulation and the other reflects our modesty manipulation. We then randomly inform participants that the CEO chose either the brag or modesty comments and ask them the extent to which "CEO Roger Smith's choice violated my expectations" on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree"). Consistent with our expectations, participants believe the CEO violated their expectations more when he is modest compared to when he brags on a conference call (50.50 > 37.20; $t_{(51)} = 1.62$, p = 0.06, one-tailed). Conversely, participants believe the CEO violated their expectations more when he brags compared to when he is modest on Twitter (50.78 > 38.97; $t_{(54)} = 1.46$, p = 0.07, one-tailed). These results support our theory and reflect that framing our question in terms of how a CEO *should* behave results in participants experiencing an expectations violation when the CEO behavior does not match their expectations.

4.6. Examining the joint effects of internal attribution and intensifying language

Our bragging manipulation is a joint manipulation, consisting of internal attribution and intensifying language. We chose this joint manipulation as a strong, externally valid, level of bragging based on theory and the results of the pilot study reported in Section 3.4. To individually examine the components of our joint manipulation, we randomly assigned participants to two additional conditions when conducting our primary experiment. In these conditions, the CEO attributes the firm's successes to the same internal attributes as the other bragging conditions, but without the intensifying language. Like other conditions, participants receive the CEO's comments either via a

[20] We ask about investors' beliefs rather than participants' personal beliefs to minimize the possibility that participants' responses are influenced by their randomly assigned experimental condition. We confirm that participants' randomly assigned condition does not influence their responses to the expectations question (all p-values > 0.21, two-tailed).

conference call transcript or via Twitter. Appendix B, Panels G and H contain images of our manipulation.

Theory suggests that intensifying language is a component of bragging and the use of it, in addition to internal attribution, is a stronger manipulation of bragging than internal attribution alone, which typically amplifies investor judgments. To examine whether this is true in our context, we first assess the degree to which participants perceived the firm's CEO to be bragging. As reported previously in Section 4.1, we ask participants: "I believe that CEO Roger Smith was bragging when he communicated about HSG's Q3 2016 results." Participants respond to this question on a 101-point scale with endpoints 0 ("Strongly Disagree") and 100 ("Strongly Agree"). Consistent with our expectations, participants in the internal attribution with intensifying language condition believe the CEO was bragging more than participants in the internal attribution only condition (59.90 > 45.30; $t_{(98)} = 3.47$, p = 0.01, one-tailed).

Next, we examine whether we find support for H1 when we manipulate bragging as only internal attribution without intensifying language. An untabulated ANOVA with change in willingness to invest as the dependent variable reveals a significant communication style by disclosure medium interaction ($F_{(1,3)} = 6.38$, p < 0.01, one-tailed), consistent with H1. Follow-up simple effects tests show that when the disclosure medium of the CEO's comments is a conference call, participants are less willing to invest when the CEO is modest compared to when the CEO brags (mean change of −2.85 < 0.77; $t_{(94)} = 1.58$, p = 0.06, one-tailed). However, when the disclosure medium of the CEO's comments is Twitter, participants are less willing to invest when the CEO brags compared to when the CEO is modest (mean change of −5.00 < -0.38; $t_{(94)} = 1.98$, p = 0.03, one-tailed). These results are consistent with H1.

Finally, we examine if internal attribution with intensifying language amplifies investor judgments compared to internal attribution alone. We find results directionally consistent with our expectations, but observed differences are not significant at conventional levels. When the disclosure medium of the CEO's comments is a conference call, the mean change in willingness to invest is 1.15 when the CEO uses internal attribution and intensifying language compared to 0.77 when the CEO uses internal attribution only ($t_{(96)} = 0.14$, p = 0.44, one-tailed). When the disclosure medium of the CEO's comments is Twitter, the mean change in willingness to invest is −6.92 when the CEO uses internal attribution and intensifying language compared to −5.00 when the CEO uses internal attribution only ($t_{(96)} = 0.71$, p = 0.24, one-tailed).

Taken together, results suggest that in both conditions we successfully manipulated bragging, but that the intensifying language we used in our brag manipulation did not have sufficient power to incrementally increase participants' judgments over internal attribution alone. In the next section, we discuss a supplemental experiment in which we increase the power of the intensifying language in our bragging manipulation to provide theory-consistent support that bragging does amplify investor judgments over internal attribution alone.

4.7. Supplemental experiment to distinguish bragging and internal attribution

In a supplemental experiment, we use a 2 × 2 between-subjects design with communication style (brag vs. internal attribution) and disclosure medium (conference call vs. Twitter) as our manipulated independent variables. To ensure we have the strongest possible manipulation of intensifying language, we first run a pilot using Amazon Mechanical Turk (MTurk) participants. In the pilot, we provide MTurk participants with the definition of intensifying language. On subsequent screens, we provide MTurk participants with variations of each sentence from our original bragging manipulation. The variations include five alternative intensifying words. MTurk participants then rate the intensity of each variation of the sentence on a 101-point scale with endpoints 0 ("Not at all Intense") and 100 ("Very Intense"). We then create a new bragging manipulation using the words rated as most intense.

Participants in our supplemental experiment are 223 graduate and senior undergraduate accounting students from a large, public university. The flow of the supplemental experiment and presentation of the materials is identical to our primary experiment except for the change in the intensifying language used in the bragging manipulation. In brag conditions, the CEO attributes the firm's successes to internal attributes and uses intensifying language as determined by the pilot. In internal attribution conditions, the CEOs comments are the same as in our primary experiment.[21] Appendix D contains images of our materials.

Similar to our primary experiment, we calculate the change in participants' willingness to invest by subtracting their final willingness to invest rating from their preliminary willingness to invest rating.[22] A Cronbach's alpha of 0.94 (0.90) for our two final (preliminary) ratings confirms that our investment-related questions are capturing the same underlying construct. Negative values indicate a decrease in willingness to invest after viewing the CEO's comments. Table 3, Panel A reports cell sizes, means, and standard deviations for the change in willingness to invest. Fig. 3 displays the pattern of cell means for our dependent variable.

Table 3, Panel B presents our ANOVA results, and Panel C presents follow-up simple effects tests. Results reveal a significant communication style by disclosure medium interaction (p = 0.03, one-tailed), consistent with our expectations. Follow-up simple effects tests reveal that when the disclosure medium of the CEO's comments is a conference call, participants are *more* willing to invest when the CEO brags (internal attribution and intensifying language) compared to when the CEO uses only internal attribution (3.22 > −0.39; t = 1.45, p = 0.07, one-tailed). When the disclosure medium of the CEO's comments is Twitter, participants are *less* willing to invest when the CEO brags (internal attribution and intensifying language) compared to when the CEO uses only internal attribution (−6.63 < −3.59; t = 1.29, p = 0.10, one-tailed).

Overall, results from our supplemental experiment confirm that the intensifying language included in the bragging manipulation in our primary experiment did not have sufficient power to incrementally amplify participants' judgments over internal attribution alone. However, when we strengthen the intensifying language, we find that bragging (internal attribution plus intensifying language) amplifies participants' judgments compared to internal attribution alone. These results are consistent with the theory we discuss in Section 2.1.

5. Conclusion and limitations

In this study, we provide evidence that a CEO's communication style interacts with disclosure medium to influence investor perceptions and investment judgments. Specifically, we find that participants who read a CEO's comments about his firm's positive performance made during a conference call (on Twitter) are *less* likely to invest in the firm when the

[21] We confirm that our communication style manipulation was successful by again comparing participants' responses across conditions to our bragging manipulation check question. Consistent with our expectations, participants in the internal attribution with intensifying language condition believe the CEO was bragging more than participants in the internal attribution only condition (59.96 > 45.46; $t_{(22)} = 4.56$, p < 0.01, one-tailed).

[22] Results are inferentially similar if we use either of the individual investment questions rather than the combined measure. Results are also robust to: 1) running an ANCOVA with the final willingness to invest as the dependent variable and preliminary willingness to invest as a covariate and 2) using final willingness to invest as the dependent variable without including preliminary willingness to invest as a covariate. Participants' willingness to invest judgments are not normally distributed in all four conditions and Levene's test for equality of variances indicates the assumption of homogeneity of variances is violated. Results are again inferentially similar using transformed values.

Table 3
Disentangling bragging and internal attribution.

Panel A: Change in Willingness to Invest, Mean (Standard Deviation), n = 223[a]

Communication Style	Disclosure Medium	
	Conference Call	Twitter
Brag	3.22	−6.63
	(11.85)	(11.02)
	n = 53	n = 58
Internal Attribution	−0.39	−3.59
	(10.92)	(16.16)
	n = 52	n = 60

Panel B: Two-Way ANOVA Model of Change in Willingness to Invest[b]

Source of Variation	Sum of Squares	df	Mean Square	F	Sig.
Communication Style	4.57	1	4.57	0.03	0.87
Disclosure Medium	2362.82	1	2362.82	14.48	0.01
Communication Style x Disclosure Medium	613.93	1	613.93	3.76	0.03[c]
Error	35725.45	219	163.13		

Panel C: Follow-Up Simple Effects Tests[d]

Source of Variation	F	Sig.
Effect of Communication Style given Conference Call	1.45	0.07[c]
Effect of Communication Style given Twitter	1.29	0.10[c]
Effect of Disclosure Medium given Brag	4.06	0.01[c]
Effect of Disclosure Medium given Internal Attribution	1.32	0.09[c]

[a] Table 3 presents tests in the supplemental experiment. Panel A presents descriptive statistics. The dependent variable is change in willingness to invest. Participants rated 1) the attractiveness of an investment in the firm's stock and 2) the likelihood that they would invest in the firm's stock. The willingness to invest variable is simply an average of these measures. Our dependent variable, change in willingness to invest, is final willingness to invest minus preliminary willingness to invest. Fig. 3 provides an illustration of these results. In the supplemental experiment, we manipulate communication style by providing comments from the CEO about the firm's performance that reflect internal attribution and intensifying language or internal attribution only. We manipulate disclosure medium by providing the CEO's comments via a conference call transcript or via Twitter.

[b] Panel B presents an ANOVA for the four conditions in the supplemental experiment.

[c] p-values are one-tailed equivalents for directional predictions.

[d] Panel C presents follow-up simple effects tests for the four supplemental conditions.

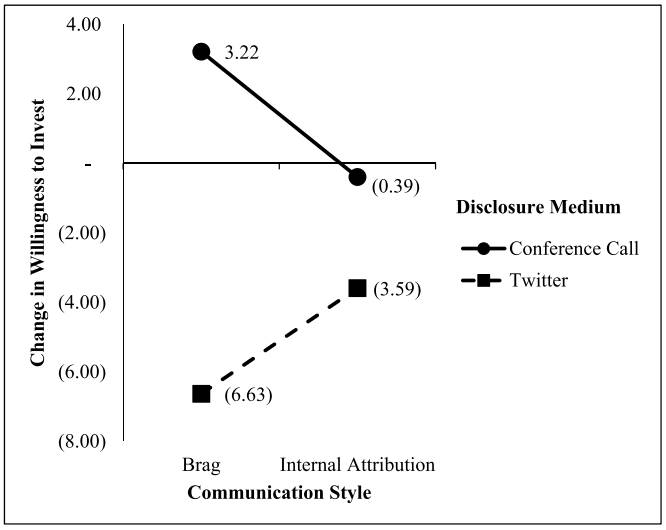


Fig. 3. Supplemental Experiment—Joint Effects of Internal Attribution and Intensifying Language on Change in Participants' Willingness to Invest.
This figure graphically depicts our observed mean values for the change in participants' willingness to invest in the supplemental experiment. Negative (positive) values indicate a decrease (increase) in willingness to invest after viewing the CEO's comments, which contain our manipulations. Table 3 presents descriptive statistics for the change in participants' willingness to invest.

CEO is modest (brags) compared to when the CEO brags (is modest). We also find, consistent with theory, that perceived CEO credibility mediates the relationship between our dependent variable (willingness to invest) and the interaction between communication style and disclosure medium. Finally, regardless of the disclosure medium of the CEO's comments, we find that participants are least willing to invest in the firm when the CEO humblebrags.

Our study has several limitations that provide opportunities for future research. First, we focus on Twitter. Though Twitter is one of the most popular social media disclosure platforms, it is not the only one. Future studies could examine if communicating with investors via other social media platforms has similar effects. Second, we operationalize the Twitter medium by abstracting away several realistic features of Twitter (e.g., the Twitter logo and bird, information about retweets, likes, etc.) to avoid a compound manipulation. Future research could examine how investors react to a Twitter manipulation containing these realistic features. Third, we use graduate and senior undergraduate students as proxies for nonprofessional investors. Future research could examine the influence of Twitter on older investors or analysts who may differ in their levels of familiarity with using social media to acquire financial information. Fourth, we examine the interaction of disclosure medium and communication style after positive firm performance. Future research could examine if our results hold when a CEO describes negative firm performance.

Fifth, we operationalize the conference call medium using a transcript rather than audio to facilitate holding information constant across conditions. Media richness theory suggests that audio is richer than a text-based transcript, which ultimately should amplify reactions to the underlying information (Daft & Lengel, 1986; Dennis & Kinney, 1998). Importantly, this theory does not predict that audio would interact with our independent variables in such a way that our inferences would change. Nevertheless, future research could examine whether our results hold when the conference call information is presented via an audio recording. Sixth, our study focuses on investors' expectations of how a CEO *should* communicate on conference calls and Twitter. Future research could examine whether these expectations differ from investors' expectations of a how a CEO *would* communicate on these disclosure mediums. Finally, we hold constant across conditions the CEO, who is unfamiliar to our investors, and examine investors' reactions to communication style. Future research could examine how over time investors might develop expectations of individual CEOs' communication styles based on their personality traits, which might interact with our independent variables to influence investors' reactions. Further, future research could examine how CEOs' personality traits might influence their communication style choices.

Overall, our study contributes to the emerging literature on social media and disclosure (e.g., Blankespoor et al., 2014; Miller & Skinner, 2015) and the literature examining style features of disclosures (e.g., Hales et al., 2011; Libby & Emett, 2014; Rennekamp, 2012; Tan et al., 2014). We also document that investors' expectations of how a CEO should communicate differ across traditional and new disclosure mediums, and show that violations of these expectations can negatively

affect investor judgments. The literature on norms examines behaviors considered acceptable in a group or society, for a particular situation (Cialdini, Kallgren, & Reno, 1991). To the extent that our manipulation of expectations captures norms, we contribute to the norms literature in accounting (e.g., Kadous & Mercer, 2011; Tayler & Bloomfield, 2011; Wenzel, 2004). Finally, our paper has practical implications for managers as they increasingly disclose information via social media outlets like Twitter. This study informs managers that the most effective communication style for conference calls is not necessarily the most effective communication style for Twitter.

Appendix A. Press release (All Conditions)

HSG Announces 2016 3rd Quarter Results

- HSG had strong top line growth with revenue of $17.986 billion, up 4.70% from the same quarter last year. The increase was driven by new marketing campaigns run by HSG and an overall increase in consumer demand for wireless products.
- Gross Margin for the quarter was 39.04%, up 1.21% from the same quarter last year.
- Earnings for the quarter were $1.028 billion, up 3.23% from the same quarter last year. Earnings per share were $1.03, beating our Q3 earnings guidance of $1.01 per share. HSG's internal teams and external consultants developed new production technologies which helped reduce the firm's manufacturing costs and increase earnings.
- R&D expenses for the quarter were $4.165 billion, down 0.74% from the same quarter last year.
- Cash Flow from Operations were $1.845 billion for the quarter, up 3.13% from the same quarter last year. Strong macroeconomic conditions and improved receivable collection practices contributed to this increase.

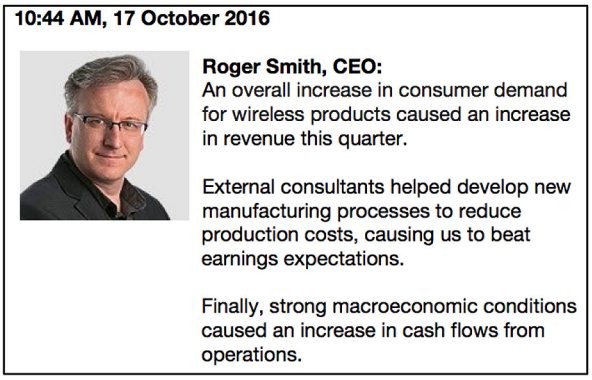
GAAP Financial Comparison			
Quarterly Year-over-Year			
	Q3 2016	Q3 2015	% change
Revenue	17,986	17,178	up 4.70%
Gross Margin	39.04%	37.83%	up 1.21 points
R&D Expenses	4,165	4,196	down 0.74%
Operating Income	1,778	1,712	up 3.86%
Net Income	1,028	991	up 3.23%
Earnings per Share	1.03	0.99	up 3.70%
Cash Flow from Operations	1,845	1,789	up 3.13%

Appendix B. Primary experiment manipulations

Panel A: CEO Brags on Conference Call



Panel B: CEO is Modest on Conference Call



Panel C: CEO Brags on Twitter



Panel D: CEO is Modest on Twitter



Appendix C. Examples of CEOs using different communication styles on Twitter

Panel A: John Legere (CEO of T-Mobile) Brags on Twitter after Q2 2017 earnings. Legere's tweet illustrates bragging because he attributes T-Mobile's positive performance to an internal factor (“Congrats Team!”) and uses intensifying language (“reigns supreme”).



Panel B: Sundar Pichai (CEO of Google) is Modest on Twitter about Chrome, an internet browser made by Google. Pichai's tweet illustrates modesty because he attributes Chrome's rise to the No. 1 spot in browser rankings to external factors ("thanks to our users and readwriteweb").



Panel C: Marcelo Claure (CEO of Sprint) Humblebrags on Twitter when he responds to an employee's tweet. Claure's tweet illustrates humblebragging because he makes a seemingly modest statement by first attributing Sprint's success to an external factor ("@Target"), but ends his tweet by promoting the employee's tweet that highlights Sprint's internal factors ("Sprint has the best looking display […] probably because we have the best promos too!").

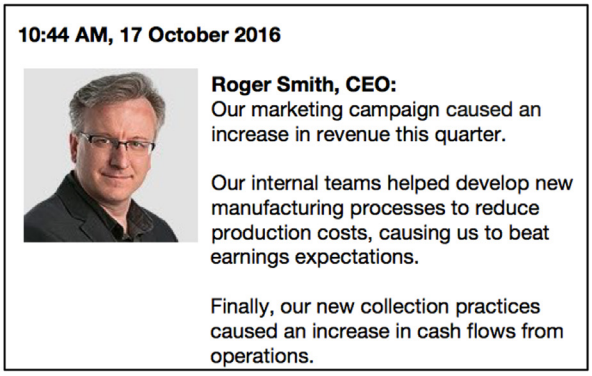


Appendix D. Supplemental experiment manipulations

Panel A: CEO Brags on Conference Call



Panel B: CEO Internally Attributes on Conference Call



Panel C: CEO Brags on Twitter



Panel D: CEO Internally Attributes on Twitter



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